PRACTICAL NO: 1A

An address in a block is given as 180.8.17.9. Find the number of addresses in the block, the first address, and the last address

Solution:

The given address is a Class B address, therefore n= 16

1) **No of addresses:**

   N = $2^{\{32-\ \}}$

   = $2^{\{32-1\ \}}$

   = $2^{\{16\}}$

   = 65536

Therefore, the number of addresses is = **65536** addresses

2) **First address:**

   For class B, address NetID = 16

   Therefore, the network mask is 255.255.0.0

   To find the first address, we logically AND the given address with the network mask

Given address	180	8	17	9
Network mask	255	255	0	0
AND operation	180	8	0	0

   Therefore, the first address is **180.8.0.0**

3) **Last address:**

   To find the last address, we logically OR the given address with the COMPLEMENT of the network mask

   Network mask = 255.255.0.0

Network mask	255	255	0	0
Complement of mask	0	0	255	255

Given address	180	8	1	9
Complement of mask	0	0	255	255
OR operation	180	8	255	255

Therefore, the last address is **180.8.255.255**

PRACTICAL NO: 1B

An organization is granted the block 130.34.12.64/26. The organization needs four subnetworks, each with an equal number of hosts. Design the sub-networks and find the information about each network.

Solution:
The given address is the address of type classless addressing with n= 26

1) **No of addresses:**
   N = $2^{\{32-n\}}$
   = $2^{\{32-26\}}$
   = $2^{\{6\}}$
   = 64

Therefore, the number of addresses is = 64 addresses

2) **First address:**
   For the given case n = 26
   Therefore, the network mask is 255.255.255.192
   To find the first address, we logically AND the given address with the network mask

Given address	130	34	12	64
Network mask	255	255	255	192
AND operation	130	34	12	64

   Therefore, the first address is 130.34.12.64

3) **Creating sub-networks:**

In this case, we need to create 4 sub-networks with an equal number of hosts

Total number of hosts N = 64

Therefore, the number of hosts in each sub-network N_{SUB}= 16

We calculate the sub-NetID for each network as follows:

$n_{SUB} = n + \log_2(N/ N_{SUB})$

$= 26 + \log_2(64/16)$

$= 28$

Therefore, the given sub-networks are:

Sub-network	First address	Last address
1	130.34.12.64	130.34.12.79
2	130.34.12.80	130.34.12.95
3	130.34.12.96	130.34.12.111
4	130.34.12.112	130.34.12.127

PRACTICAL NO: 2

Static Routing

1. The static routing method is most trusted by a router.
2. Static routing is not really a routing protocol.
3. Static routes do not dynamically adapt to network changes, are not particularly scalable, and require manual updating to reflect changes.

Static routing has the following advantages

1. There is no bandwidth usage between routers, which means you could possibly save money on WAN links.
2. There is no overhead on the router CPU, which means you could possibly buy a cheaper router than you would use if you were using dynamic routing.
3. It adds security because the administrator can choose to allow routing access to certain networks only.

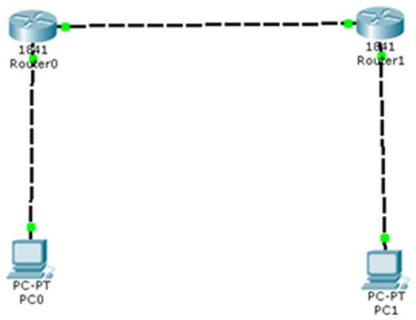
Static routing has the following disadvantages

1. Static routes don't dynamically adapt to network changes.
2. If a network is added to the internetwork, the administrator has to add a route to it on all routers—by hand.
3. It's not feasible in large networks because maintaining it would be a full-time job in itself.
4. With static routing, as your network grows, it can be difficult just keep adding static routes to make sure everybody can still get everything.
5. The administrator must really understand the internetwork and how each router is connected in order to configure routes correctly.

There are two different styles to configure an "IP route" command:

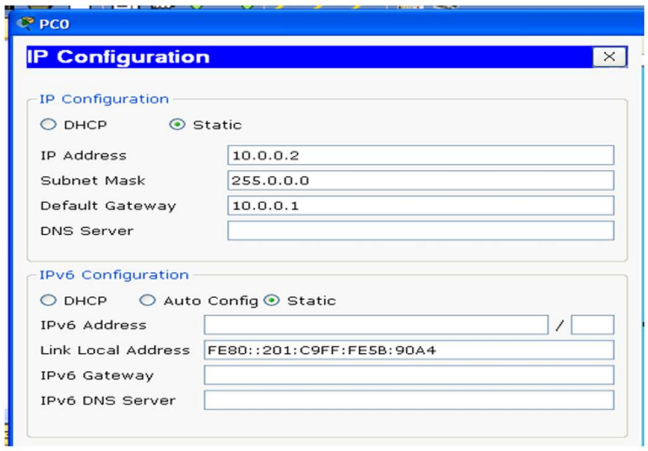
1. **Using a next hop IP address**
2. **Using an outgoing interface**

Consider the following network:

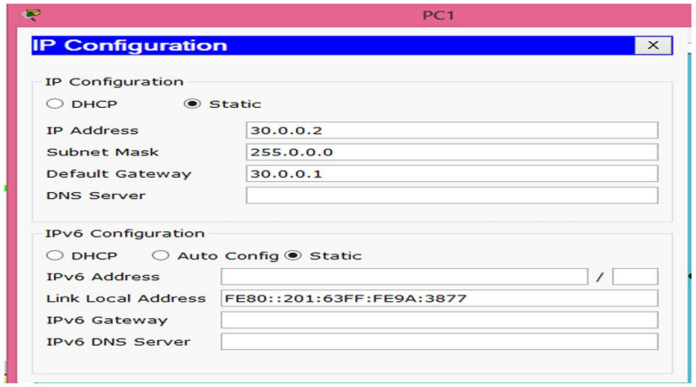


We configure it as follows:

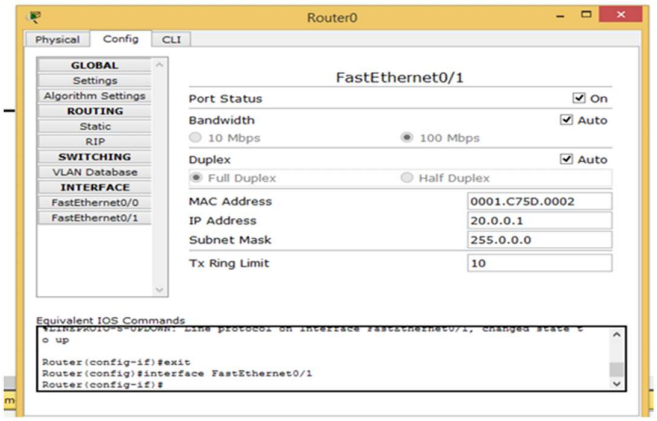
Step 1: (configure PC 0)

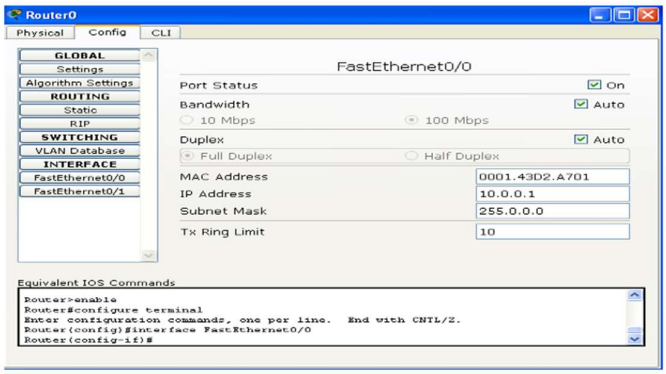


Step 2: (configure PC 1)

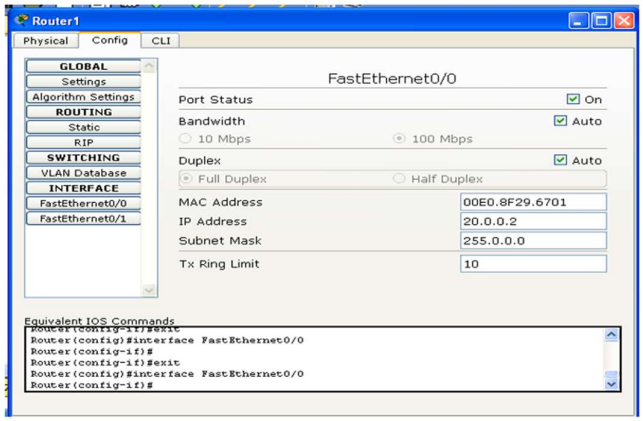


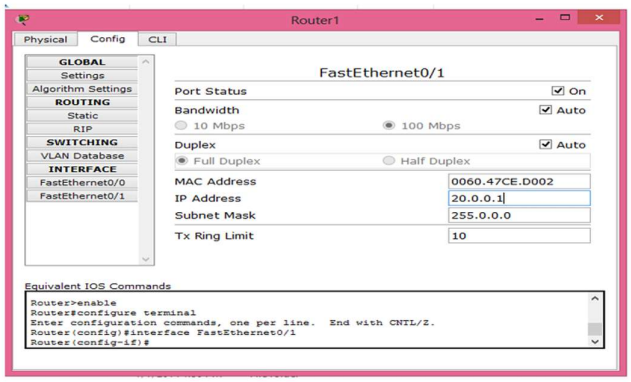
Step 3: (configure Router 0)





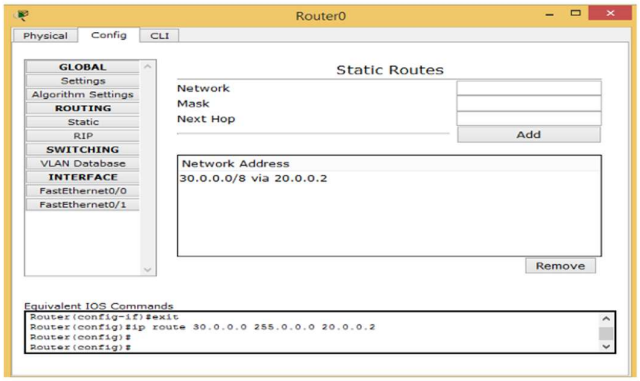
Step 4: (configure Router 1)



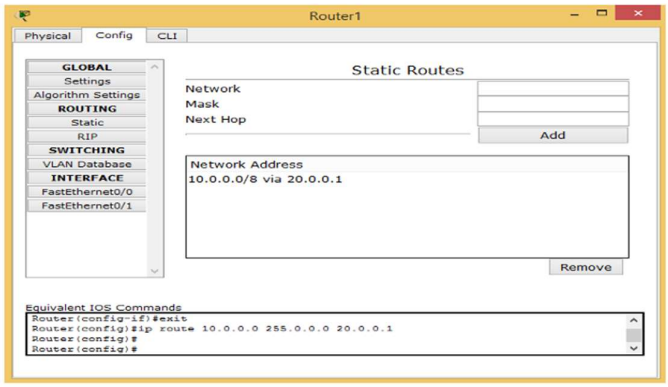


The routing table is configured in the following way

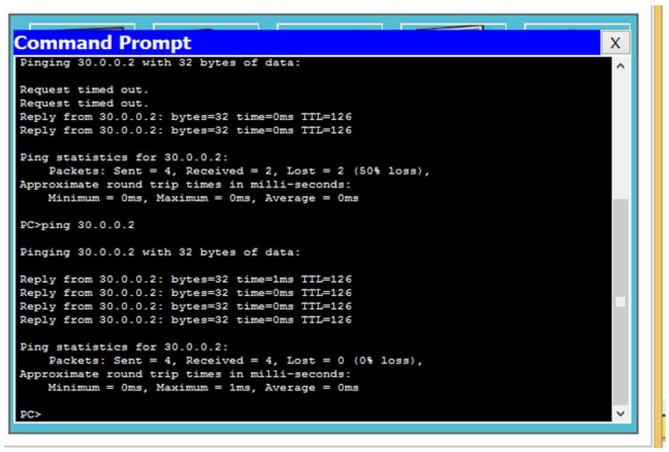
For router 0



For router 1



Now we can give the ping command, as shown, to check the connectivity



So static routing has been studied

PRACTICAL NO: 3

RIP PROTOCOL DEMONSTRATION USING CISCO PACKET TRACER

AIM:

To configure and demonstrate the Routing Information Protocol (RIP) using Cisco Packet Tracer and verify communication between different networks.

APPARATUS / SOFTWARE REQUIRED

- Cisco Packet Tracer
- 3 Routers (2911)
- 3 PCs
- Copper Straight-Through Cables

THEORY

Routing Information Protocol (RIP) is a dynamic routing protocol that uses hop count as its metric to determine the best path to a destination network.

RIP version 2 (RIPv2) supports classless routing, subnet masks, and multicast updates.

- Maximum hop count: 15
- Routing updates every 30 seconds
- Uses UDP port 520

NETWORK TOPOLOGY

PC1 ---- R1 ---- R2 ---- R3 ---- PC3

IP ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask
PC1	NIC	192.168.1.10	255.255.255.0
R1	G0/0	192.168.1.1	255.255.255.0
R1	G0/1	10.0.0.1	255.255.255.0
R2	G0/0	10.0.0.2	255.255.255.0
R2	G0/1	20.0.0.1	255.255.255.0
R3	G0/0	20.0.0.2	255.255.255.0
R3	G0/1	192.168.2.1	255.255.255.0

PC3	NIC	192.168.2.10	255.255.255.0

PROCEDURE:

Step 1: Create Network Topology

1. Open Cisco Packet Tracer
2. Place three routers and three PCs
3. Connect all devices using straight-through cables

Step 2: Configure Router Interfaces

Router R1

```
enable
configure terminal
interface g0/0
IP address 192.168.1.1 255.255.255.0
no shutdown
exit
interface g0/1
IP address 10.0.0.1 255.255.255.0
no shutdown
exit
```

Router R2

```
enable
configure terminal
interface g0/0
IP address 10.0.0.2 255.255.255.0
no shutdown
exit
interface g0/1
IP address 20.0.0.1 255.255.255.0
no shutdown
exit
```

Router R3

```
enable
configure terminal
interface g0/0
IP address 20.0.0.2 255.255.255.0
no shutdown
exit
```

```
interface g0/1
IP address 192.168.2.1 255.255.255.0
no shutdown
exit
```

Step 3: Configure RIP Routing

On Router R1

```
router rip
version 2
network 192.168.1.0
network 10.0.0.0
no auto-summary
```

On Router R2

```
router rip
version 2
network 10.0.0.0
network 20.0.0.0
no auto-summary
```

On Router R3

```
router rip
version 2
network 20.0.0.0
network 192.168.2.0
no auto-summary
```

Step 4: Configure PCs

PC1

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

PC3

- IP Address: 192.168.2.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

VERIFICATION

From PC1, open Command Prompt:

```
ping 192.168.2.10
```

A successful reply confirms RIP routing is working.

Use the command on routers:

```
show IP route
```

RESULT

RIP protocol was successfully configured, and communication between different networks was verified.

PRACTICAL NO: 5

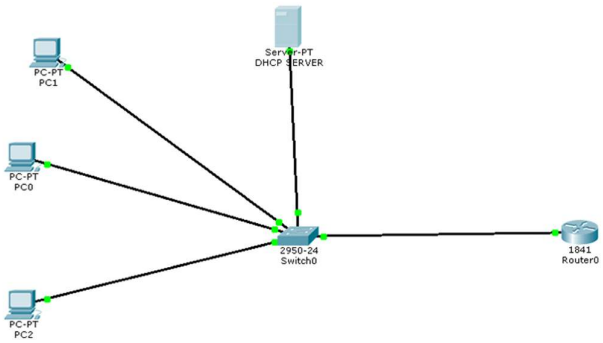
Configuring DHCP and DNS

DHCP

The Dynamic Host Configuration Protocol (DHCP) is a client/server protocol designed to provide the four pieces of information for a diskless computer or a computer that is booted for the first time. DHCP is a successor to BOOTP and is backwards compatible with it. Although BOOTP is considered deprecated, there may be some systems that still use BOOTP for host configuration. The part of the discussion in this chapter that does not deal with the dynamic aspect of DHCP can also be applied to BOOTP. The DHCP client and server can either be on the same network or on different networks. When on the same network, it works as follows.

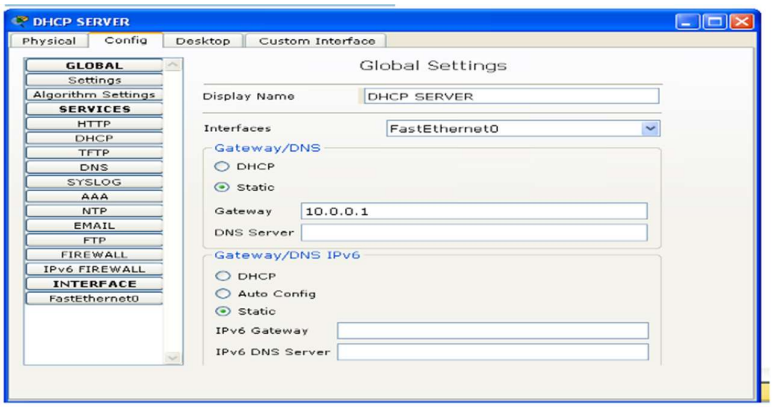
1. The DHCP server issues a passive open command on UDP port number 67 and waits for a client.
2. A booted client issues an active open command on port number 68. The message is encapsulated in a UDP user datagram, using the destination port number 67 and the source port number 68. The UDP user datagram, in turn, is encapsulated in an IP datagram. The reader may ask how a client can send an IP datagram when it knows neither its own IP address (the source address) nor the server's IP address (the destination address). The client uses all 0s as the source address and all 1s as the destination address.
3. The server responds with either a broadcast or a unicast message using UDP source port number 67 and destination port number 68. The response can be unicast because the server knows the IP address of the client. It also knows the physical address of the client, which means it does not need the services of ARP for logical to physical address mapping. However, some systems do not allow the bypassing of ARP, resulting in the use of the broadcast address

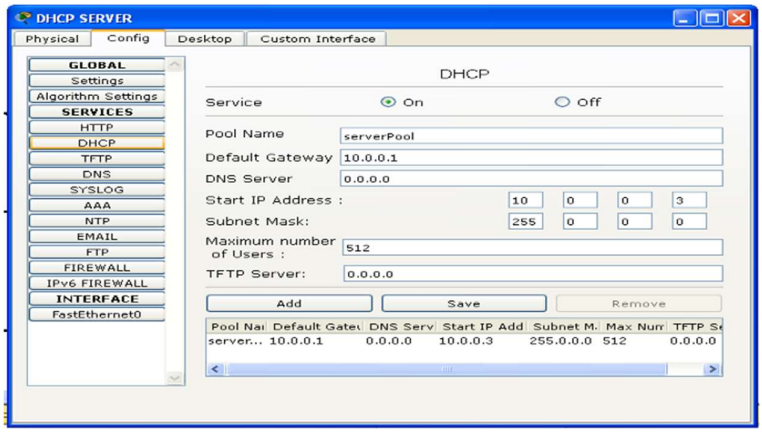
We can study the working of DHCP using the cisco packet tracer using the following example.



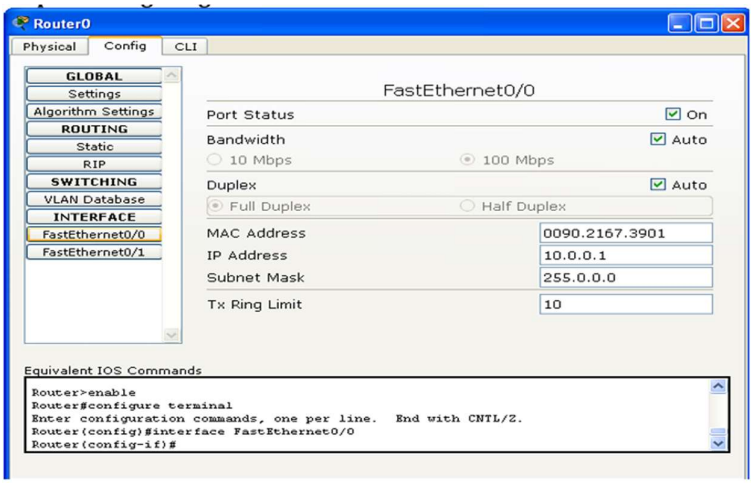
We configure the various components through the following steps:

Step 1: Configuring the DHCP server



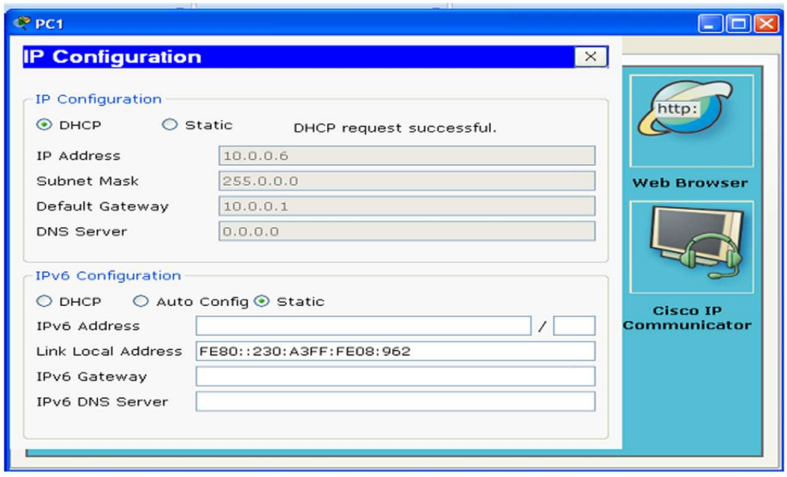


Step 2: Configuring the Router



Now we test the working of the DHCP server by sending a DHCP request from any of the PC as shown

Step 3: Sending DHCP request



Hence, we have configured a DHCP server and also verified its operation

DNS

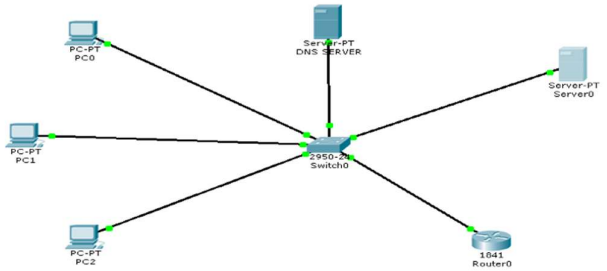
DNS To identify an entity, TCP/IP protocols use the IP address, which uniquely identifies the connection of a host to the Internet. However, people prefer to use names instead of numeric addresses. Therefore, we need a system that can map a name to an address or an address to a name. One of the solutions is to divide this huge amount of information into smaller parts and store each part on a different computer. In this method, the host that needs mapping can contact the closest computer holding the needed information. This method is used by the **Domain Name System (DNS)**

The DNS consists of the following steps

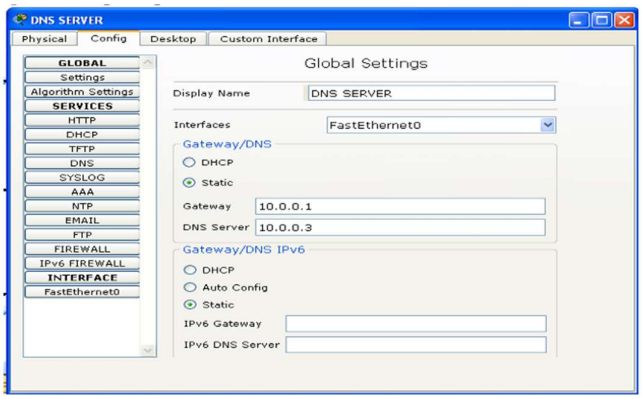
1. The user passes the host name to the file transfer client.

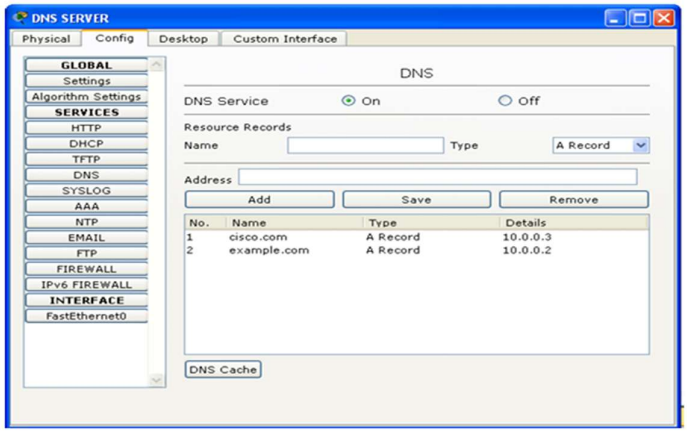
2. The file transfer client passes the hostname to the DNS client.
3. The DNS client sends a message to a DNS server with a query that gives the file transfer server name using the known IP address of the DNS server.
4. The DNS server responds with the IP address of the desired file transfer server.
5. The DNS client passes the IP address to the file transfer server.
6. The file transfer client now uses the received IP address to access the file transfer server.

We can study the working of DNS using the cisco packet tracer using the following example.

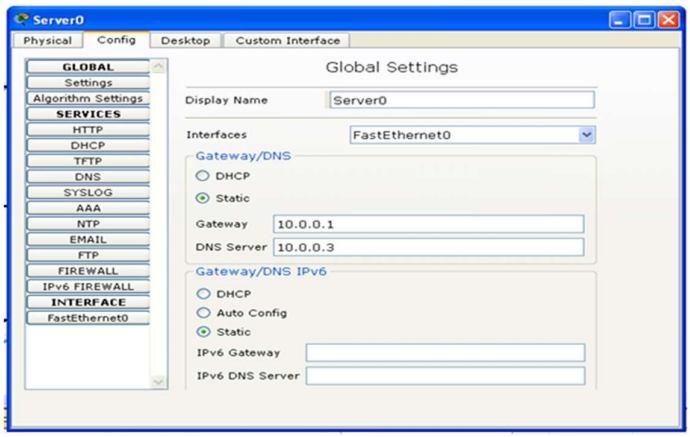


Step 1: Configuring the DNS server

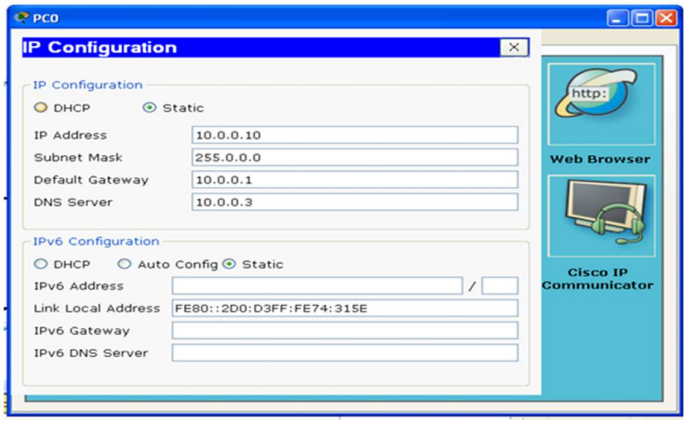




Step 2: Configuring the server 0

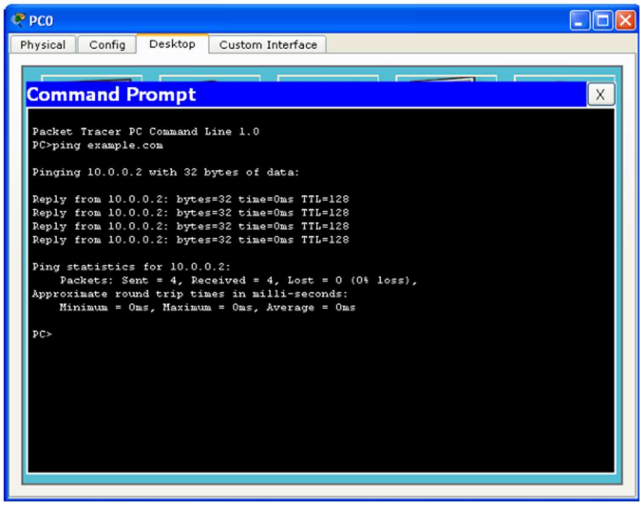


Step 3: Configuring PC0



Similarly, we can assign any IP address to PC1 and PC2

Now we can verify the working as follows



Hence, the working of DNS has been studied